

ONLINE RETAIL SALES DATABASE DESIGN

Introduction

The Online Retail Sales Database Design project is developed to manage and organize the data of an e-commerce platform. The database stores important information related to customers, products, orders, order items, payments, and inventory transactions.

The main objective of this project is to design a structured and normalized relational database that maintains data consistency and reduces redundancy. An Entity Relationship Diagram (ERD) is also designed using dbdiagram.io to represent the entities and their relationships.

The project also includes SQL queries and views that can be used to analyze sales, customer spending, product performance, and inventory information.

Abstract

This project implements an Online Retail Sales Database using MySQL. The database consists of six major entities: Customers, Products, Orders, Order Items, Payments, and Inventory Transactions.

The database is designed using 3NF normalization principles. Primary keys and foreign keys are used to establish relationships between tables, while constraints such as UNIQUE, NOT NULL, DEFAULT, and CHECK are used to maintain data integrity.

Sample data is inserted to represent a realistic online retail environment. SQL JOINS, aggregate functions, views, and the RANK () window function are used to generate useful business reports.

An additional inventory management feature was implemented using an inventory transaction table and a trigger that automatically updates product stock when inventory transactions are recorded.

Tools Used

Tool	Purpose
MySQL	Database creation and SQL implementation
MySQL Workbench	Writing and executing SQL queries
dbdiagram.io	Designing the ER diagram
SQL	Tables, constraints, joins, views, triggers and reports

Steps Involved in Building the Project

Step 1: Database and Schema Design

The database online_retail_db was created and the main entities were identified: Customers, Products, Orders, Order Items, Payments, and Inventory Transactions.

Step 2: ER Diagram

An Entity Relationship Diagram (ERD) was designed using dbdiagram.io to show the tables, primary keys, foreign keys, and relationships between entities.

Step 3: Normalization

The database was designed according to Third Normal Form (3NF) to reduce data redundancy and improve data consistency.

Step 4: Table Creation and Constraints

SQL CREATE TABLE statements were used to create the tables. Primary keys, foreign keys, UNIQUE, NOT NULL, DEFAULT, and CHECK constraints were implemented to maintain data integrity.

Step 5: Sample Data

Sample records were inserted into the customer, product, order, order-item, payment, and inventory transaction tables.

Step 6: SQL Queries and Views

SQL JOINS, aggregate functions, and views were created to generate sales and customer reports. Reports include product sales, category sales, customer spending, and order details.

Project Improvements

The basic project was enhanced with additional SQL features:

- **Inventory Transactions:** Records purchases, sales, and returns for products.
- **Automatic Stock Update Trigger:** Automatically updates product stock when a new inventory transaction is recorded.
- **Low Stock View:** Identifies products as IN STOCK, LOW STOCK, or REORDER NOW.
- **Product Sales Ranking:** Uses the RANK() window function to rank products based on total sales.
- **Query Performance Analysis:** Uses EXPLAIN to examine query execution and performance.

These improvements make the database more practical for an e-commerce environment and demonstrate additional SQL skills.

Conclusion

The Online Retail Sales Database successfully demonstrates the design and implementation of a normalized relational database for an e-commerce platform. The project covers ER modeling, 3NF normalization, table creation, constraints, sample data, JOIN queries, aggregate functions, views, triggers, window functions, and query performance analysis.

The additional inventory management and automated stock-update features extend the basic database functionality and provide a more practical solution for managing retail operations.